

Basic Principles of Medicine 2

Module : Digestion and Metabolism

DLA before Lecture 34

Obesity

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Acknowledgement
Dr. Sharmila Upadhya

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Objectives: Obesity

- After DMLecture 34

SOM. MK.III.BPM2.3.DM.3.BCHM.1328	Define obesity using body mass index (BMI)
SOM. MK.III.BPM2.3.DM.3.BCHM.1329	Compare the various methods available for diagnosis of obesity including BMI, waist circumference, Waist to hip ratio, bioelectric impedance and discuss the significance of location of body fat ('apple' vs 'pear') in obesity.
SOM. MK.III.BPM2.3.DM.3.BCHM.1330	Summarize factors contributing (multifactorial) to obesity: including genetic, environmental, behavioral factors and the role of leptin.
SOM. MK.III.BPM2.3.DM.3.BCHM.1331	Analyze metabolic changes in obesity: including Dyslipidemia (Correlate changes in lipid profile to metabolic changes in obesity) and Syndrome X or Metabolic syndrome (Correlate clinical and biochemical features and explain significance of insulin resistance).
SOM. MK.III.BPM2.3.DM.3.BCHM.1332	Explain nutritional and lifestyle changes advised for the management of obesity.
SOM. MK.III.BPM2.3.DM.3.BCHM.1333	Evaluate the role of surgery and drugs (sibutramine and orlistat) in obesity management.

Assigned Reading

- **Lippincott Illustrated Reviews, Biochemistry, 9th Edition:**

 <https://sgu.idm.oclc.org/login?url=https://premiumbasicsciences.lwwhealthlibrary.com/book.aspx?bookid=3402>

- **Lippincott's Biochemistry:**

- **Chapter 26:** [Obesity | Lippincott Illustrated Reviews: Biochemistry, 9e | Premium Basic Sciences | Health Library](#)

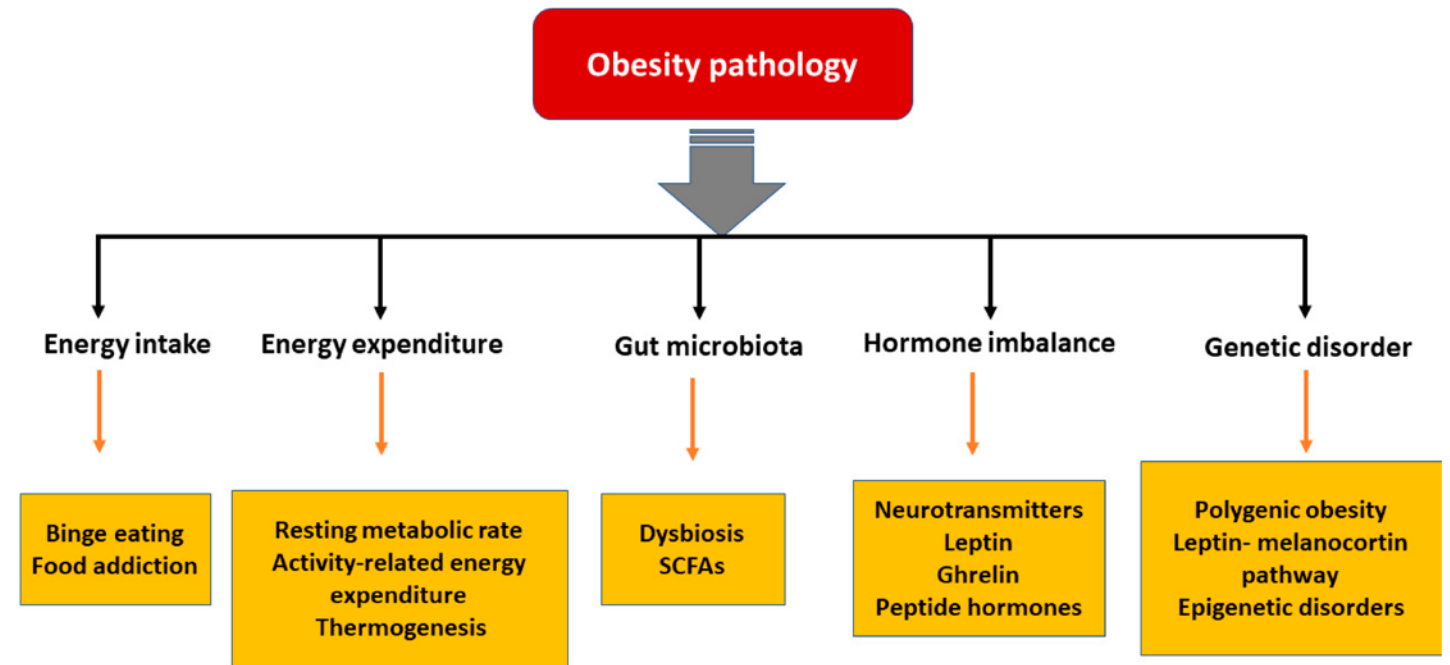
- Question: 26.1 and 26.2
- Concept map: p.355

- **Rubin's Pathology, Chapter 13:** Obesity and Diabetes mellitus:

<https://premiumbasicsciences.lwwhealthlibrary.com/book.aspx?bookid=1040>

Obesity

- **Obesity:** Excess or abnormal fat accumulation that presents a risk to health.
- Due to a long-standing **caloric imbalance** between **energy intake** and **energy expenditure**, including **several factors that favor intake**



<https://doi.org/10.3390/nu13020510>

- Global epidemic: Affects adults, adolescents, and children
- **Primary Obesity:** Due to caloric imbalance.
- **Secondary Obesity:** Endocrine disorders (e.g., Cushing's syndrome, insulinoma), genetic syndromes (e.g., Prader-Willi syndrome), medications (e.g., antidepressants, steroids ...)

Obesity and Body Mass Index (BMI)

- Amount of body fat determined indirectly by estimating body mass index (BMI).
- BMI is measure of relative weight, adjusted for height.

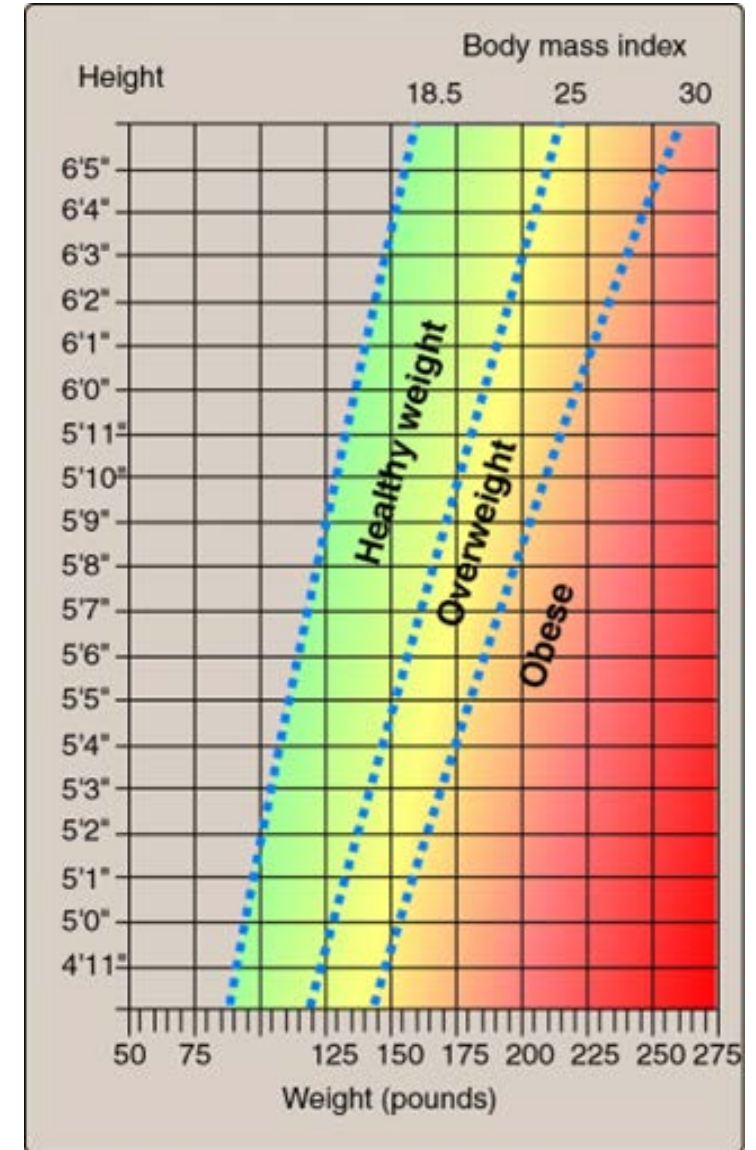
$$\text{BMI} = \frac{\text{Weight (kg)}}{\text{Height (m}^2\text{)}}$$

Adult BMI Reference Ranges kg/m²

- **Healthy range** = 18.5 - 24.9
- **Overweight range** = 25 - 29.9
- **Obese** ≥ 30

[Calculate Your BMI - Standard BMI Calculator](#)

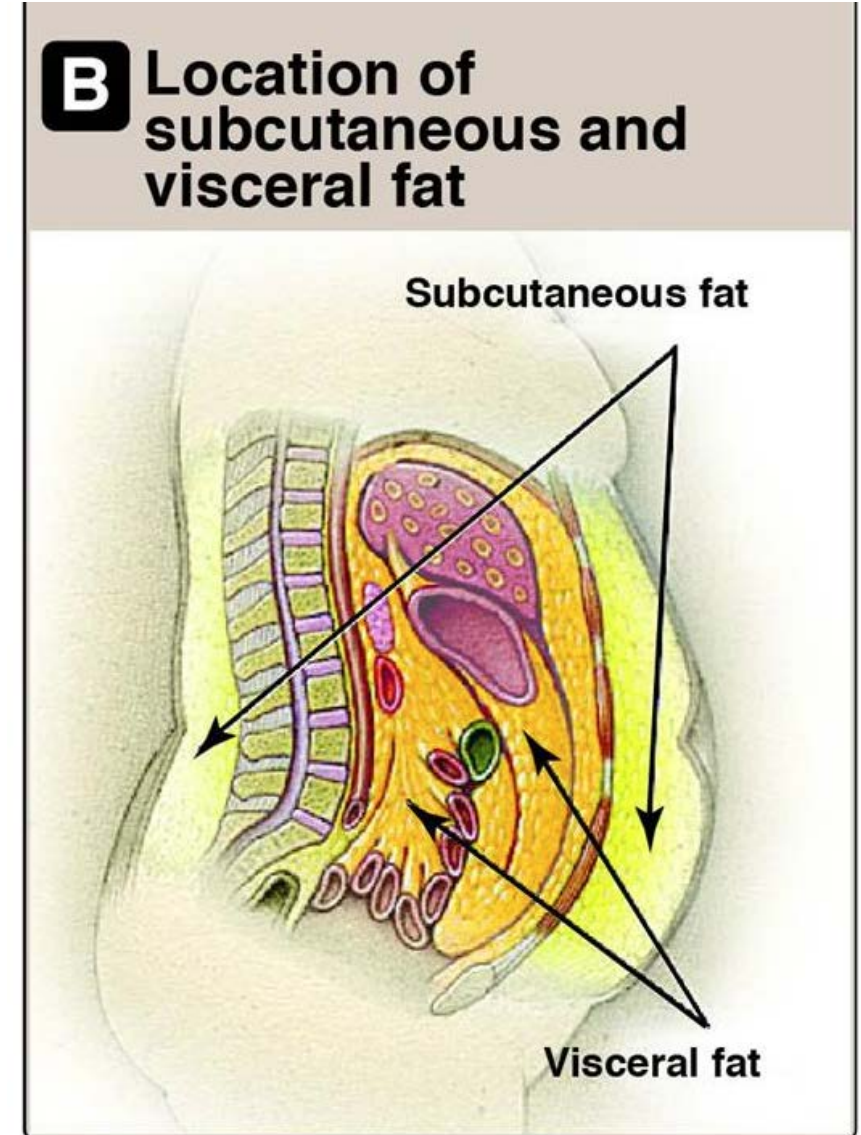
- Strongly correlates with total fat content in adults
- Exceptions: Athletes who have higher lean muscle mass



BMI Chart: height/weight in feet/pounds

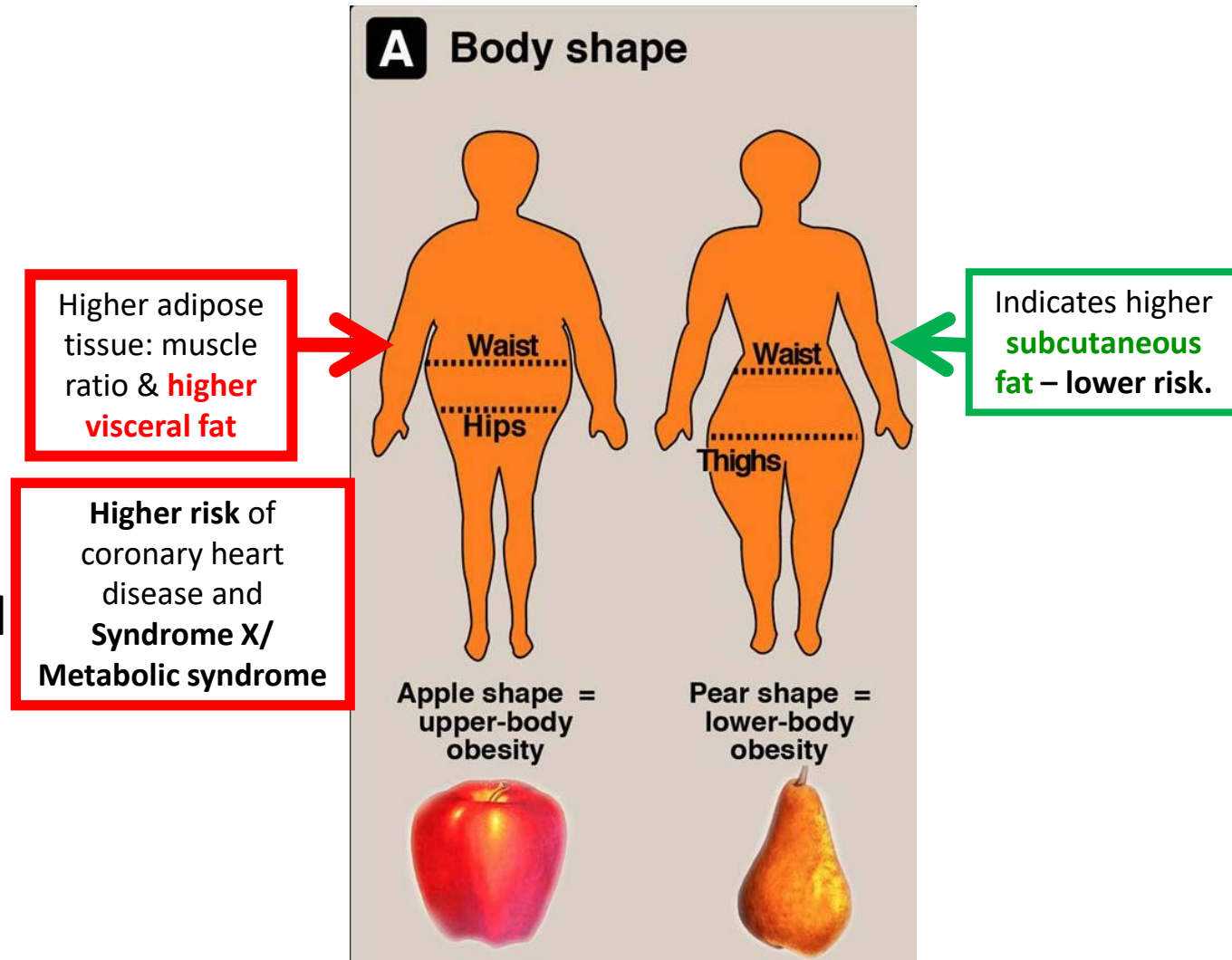
Body Fat Distribution

- Two kinds of fats based on location:
 - **Visceral** and
 - **Subcutaneous**
- **Visceral** or abdominal fat: Fatty acids and cytokines are released into **portal vein**, taken up by liver and contributes to **higher VLDL** levels
 - **Higher Cardiometabolic risk**
- **Subcutaneous** fat: Fatty acids released enter **general circulation** and used by muscle before reaching liver
 - **Lower health risk**



Anatomic Differences in Fat Deposition

- **Android, apple-shaped, or upper-body obesity or central obesity or abdominal obesity** - more fat deposition in trunk, mesentery (**visceral**)
 - Waist circumference >35 in for women and >40 in for men
 - **Higher risk of cardiometabolic disease**
- **Gynoid, pear-shaped, or lower-body obesity** - lower WHR reflects fat distributed in the hips and thighs (**subcutaneous**)
 - **Lower risk of metabolic disease**
- Body shape indices used to identify higher risk of obesity-associated metabolic disease



Persons with **upper-body obesity** (left) have greater health risks than persons with **lower-body obesity** (right).

Diagnosing Obesity

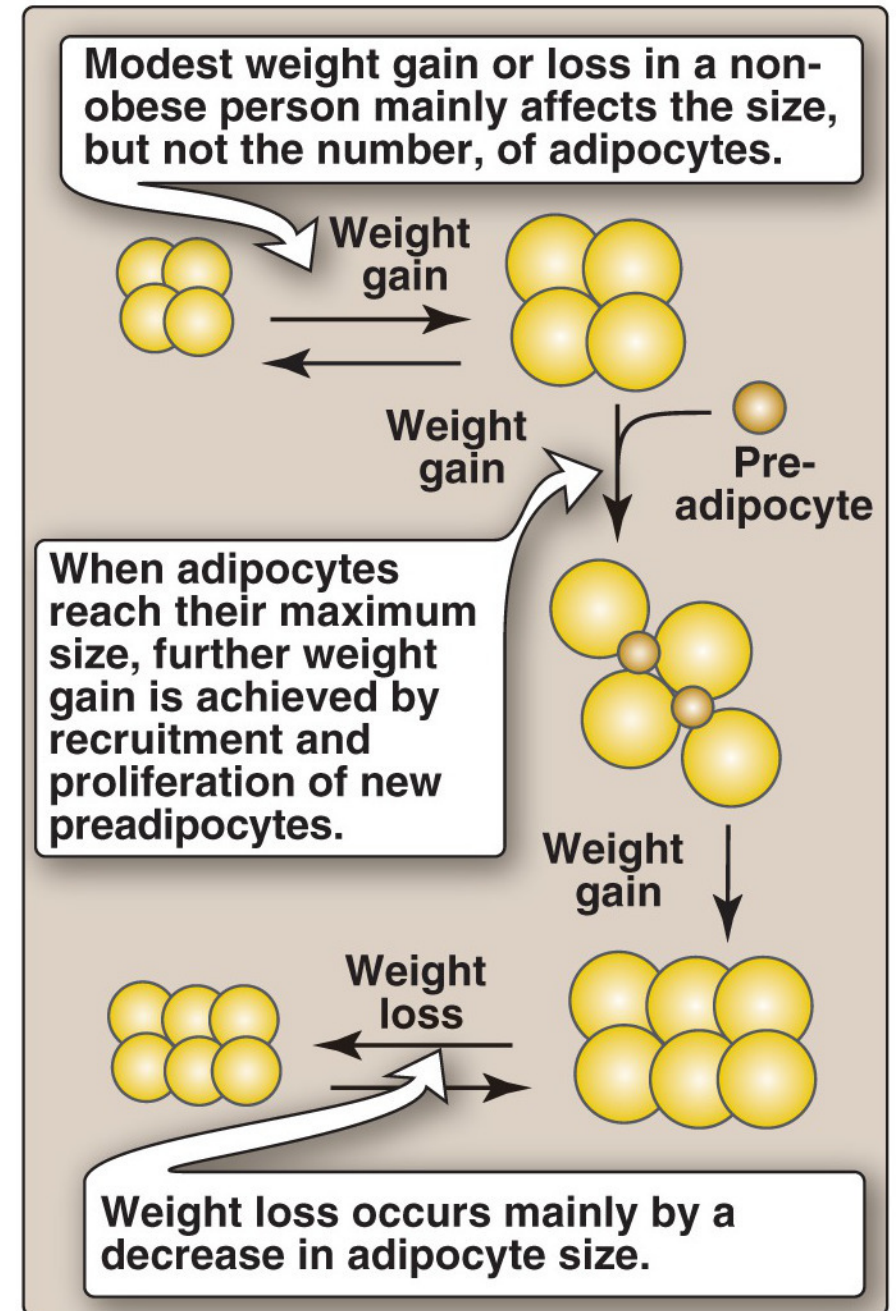
- **Appearance** - visual
- **BMI** calculation
- **Waist circumference**: ≥ 40 inches for men; ≥ 35 inches in women
 - Higher waist circumference increases health risks (men and women)
- **Waist to hip ratio** (WHR): ≥ 0.9 for men; ≥ 0.8 for women (Used less frequently)

Methods to Assess Body Fat Content

- **Skin fold measurements**: Assess subcutaneous fat and compare to population standards
- Bioelectric impedance
- Dual-energy x-ray absorptiometry (DEXA)
- Underwater weighing

Adipocyte Changes in Obesity

- Triglycerides (TAG) constitute nearly 90% of adipocyte mass
- Modest weight gain results in an increase in adipocyte size (**hypertrophy**)
- Excessive weight gain (**obesity**) results in increase in both size (**hypertrophy**) and number (**hyperplasia**) of adipocytes
- Weight loss results in a decrease in size of adipocytes



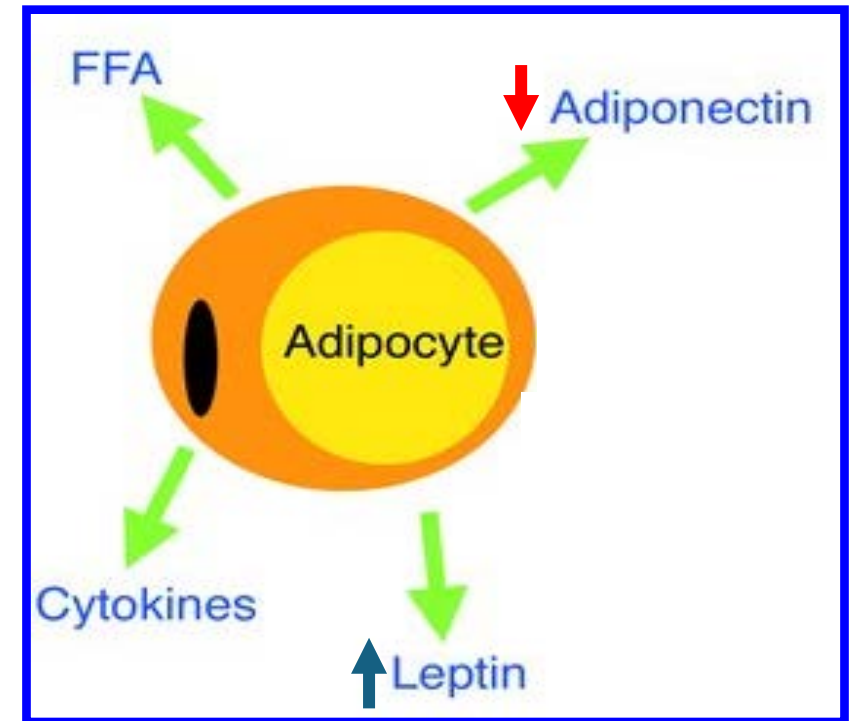
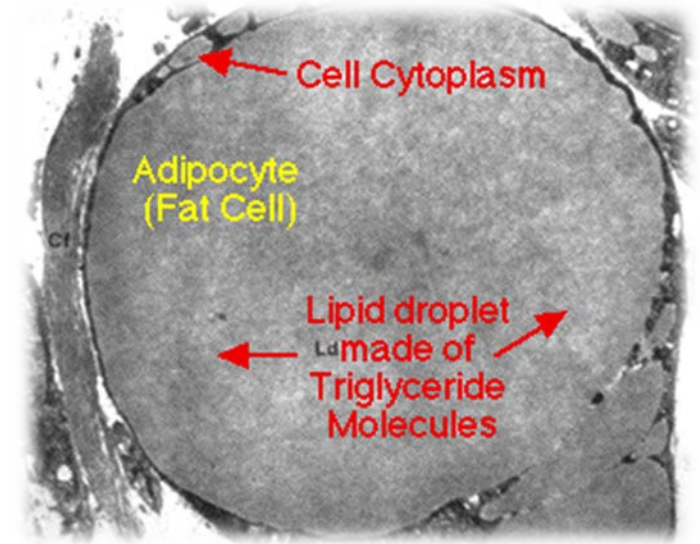
Changes in Adipocytes Endocrine Function in Obesity

- Adipocytes secrete **hormones** that regulate:
 - Appetite control/weight gain (**leptin**-appetite suppressant)
 - Inflammation (**cytokines** & **adiponectin**)
 - Insulin resistance (**adiponectin**)

- **Weight gain: Hormone changes**

- Increases **leptin** secretion
- Decreases **adiponectin** secretion

Leptin resistance: Most obese individuals have elevated leptin levels but resistant to appetite-regulating effects of leptin.



Obesity: Multifactorial Disorder: Genetics and Environment

- **Multiple genetic factors:** Obesity seen in leptin-melanocortin gene mutations, and epigenetic disorders like Prader-Willi syndrome
- **Genes set the stage.** Polygenic; However, weight determined by **diet & exercise (activity)**
- **Complex underlying contributors:** Include interaction of Genetic, Environmental, Behavioral, Metabolic, Physiologic, and Social factors.



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Identical twins (raised together or apart) with similar BMI's

Genetic Components of Obesity

- **Monogenic Mutations linked to obesity** are shown in table
- **Mutations** may result in hyperphagia and severe obesity

Name	Gene	Function
Leptin	<i>LEP</i>	Secreted by adipocytes and functions as satiety signal in hypothalamus
Leptin receptor	<i>LEPR</i>	Functions as a receptor for leptin to mediate its effect
Proopiomelanocortin	<i>POMC</i>	Its deficiency results in the absence α MSH which regulates appetite
Melanocortin 4 receptor	<i>MC4R</i>	Appetite regulation, binds to α MSH

<https://doi.org/10.3390/nu13020510>



A: Patient with leptin deficiency before initiation of therapy at age 5-years. **B:** Patient at age 9-years after 48-months of therapy with subcutaneous injections of recombinant leptin (adapted from Lippincott Illustrated Reviews, Biochemistry 8 ed.).

Environmental and Behavioral Factors in Obesity

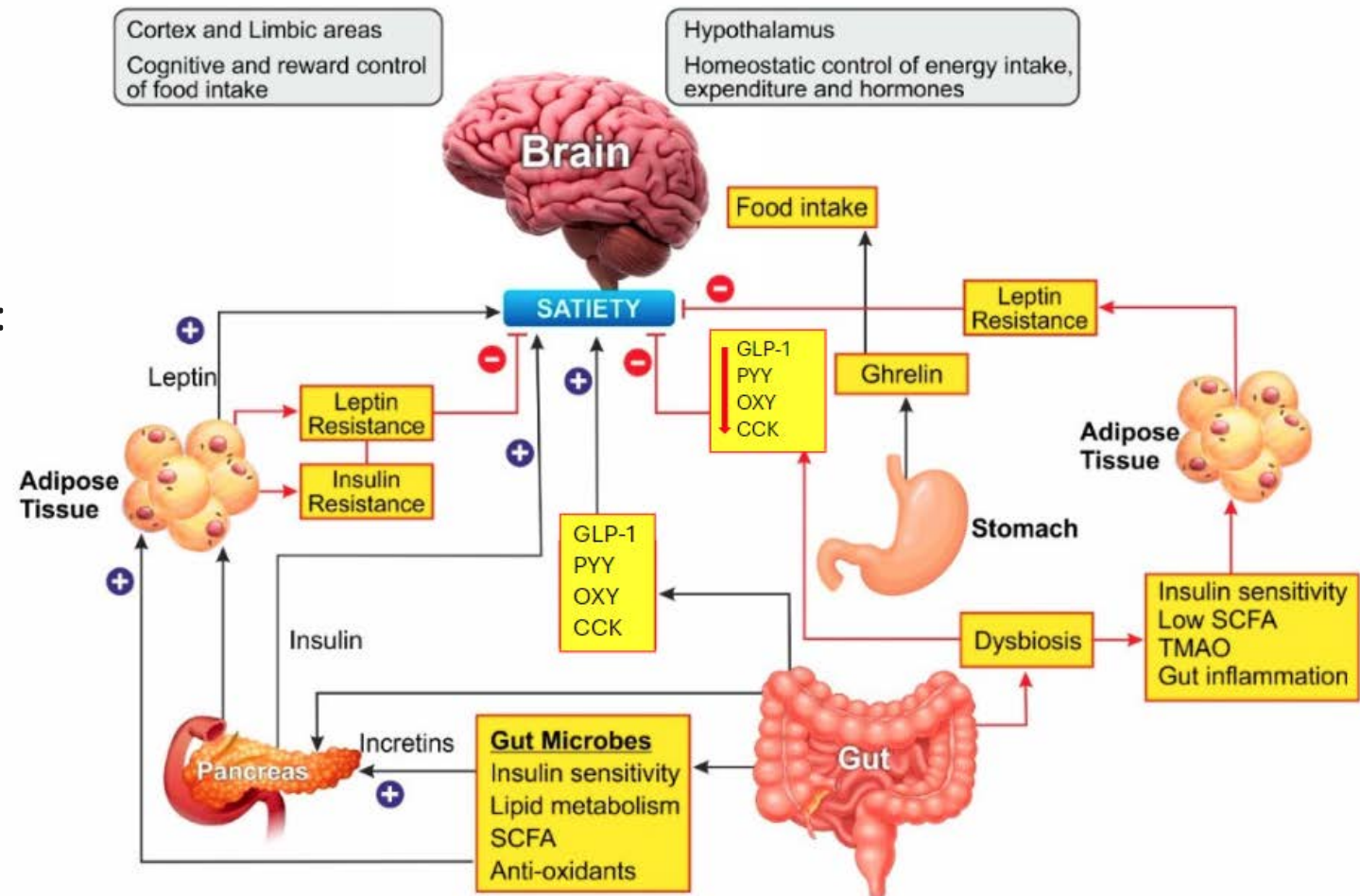
- Availability of palatable, energy-dense foods
- Sedentary lifestyles: Tendency to gain weight.
- Eating behaviors: Portion size, variety of foods consumed, food preferences, and number of people present during eating.
 - Many individuals in the same environment do not become obese.
- Susceptibility to obesity: **Interaction of genes and environment** influenced by additional factors:
 - Maternal under- or overnutrition during pregnancy
 - Epigenetic changes influence risk of obesity



Increased obesity rates indicate that the fundamental problem is most likely recent changes in **environmental and behavioral factors**, not ruling out the contribution of **genetics**.

Regulation of Obesity by Hormones and Gut Microbiome

- **Brain** (central control of body weight) receives hormonal and neuronal signals about food deficit or surplus and fuel utilization rate
- **Ghrelin** (stomach): Appetite stimulator (**Hunger signal**)
- **Insulin** (pancreas) and **leptin** (adipose tissue): Induce satiety (**Fullness signal**)
- Gut-derived peptides (**GLP-1, PYY, OXY, CCK**) augment insulin release and induce satiety
- **Gut microbiome** balances energy homeostasis. Releases short-chain fatty acids (SCFA) with numerous benefits
- **Dysbiosis** of gut:
 - Decreases peptide hormone and
 - Increases gut inflammation
 - Induce insulin and leptin resistance preventing satiety



Adapted from: <https://doi.org/10.3390/nu13020510>

CCK, cholecystokinin; GLP-1, glucagon-like peptide-1; OXY, oxyntomodulin; PYY, peptide YY; SCFA, short-chain fatty acids. Black lines represent positive stimulation to control fat storage while red lines represent the causes of obesity.

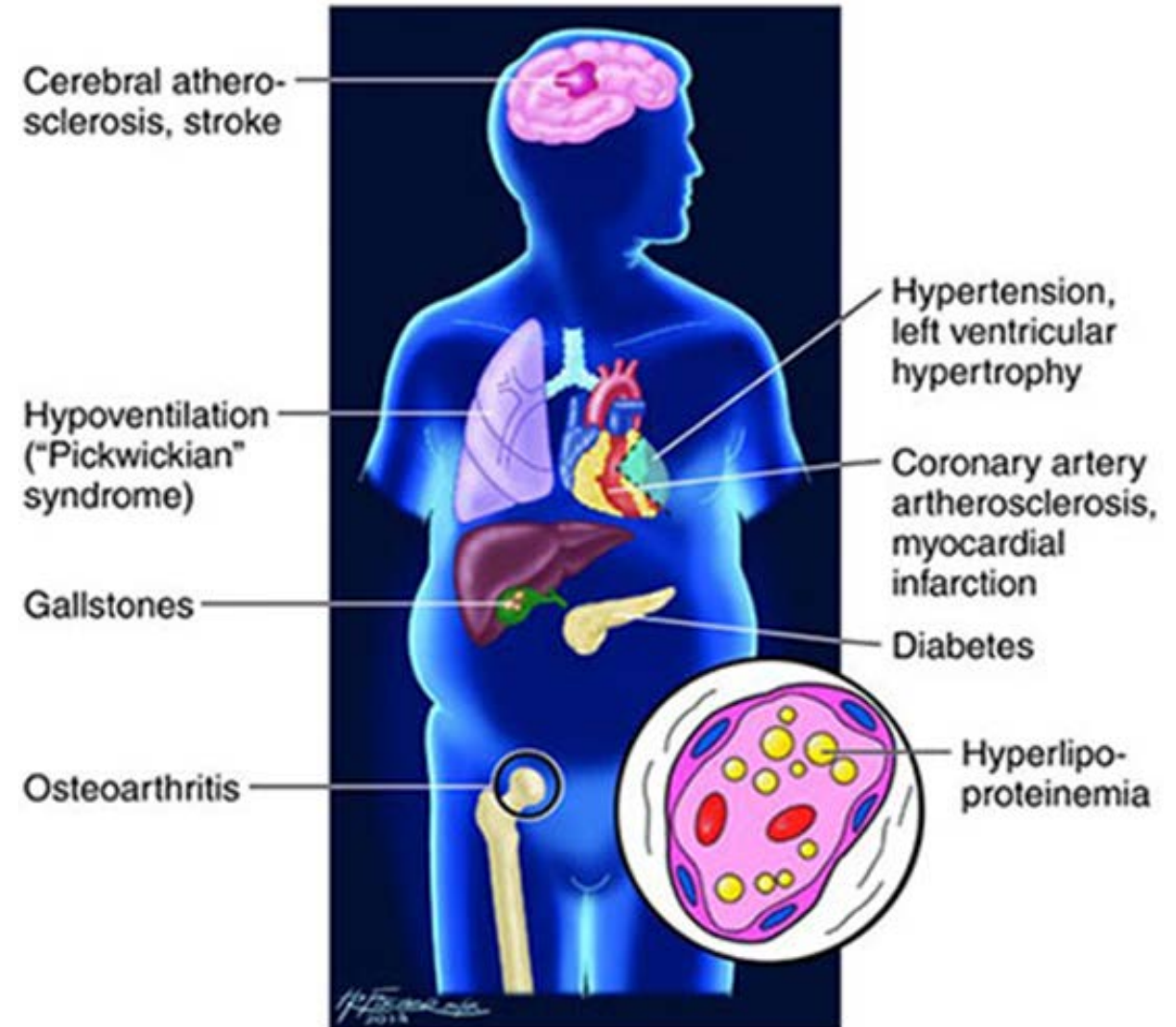
Metabolic effects of Obesity

Metabolic effects

- Dyslipidemia (increase in serum TAGs especially VLDL)
- Glucose intolerance
- Insulin resistance syndrome

Increased risk of:

- Diabetes mellitus type 2
- Hypertension
- Atherosclerosis
- Cardiovascular disease
- Stroke
- Non-alcoholic fatty liver disease
- Obstructive sleep apnea
- Cancer

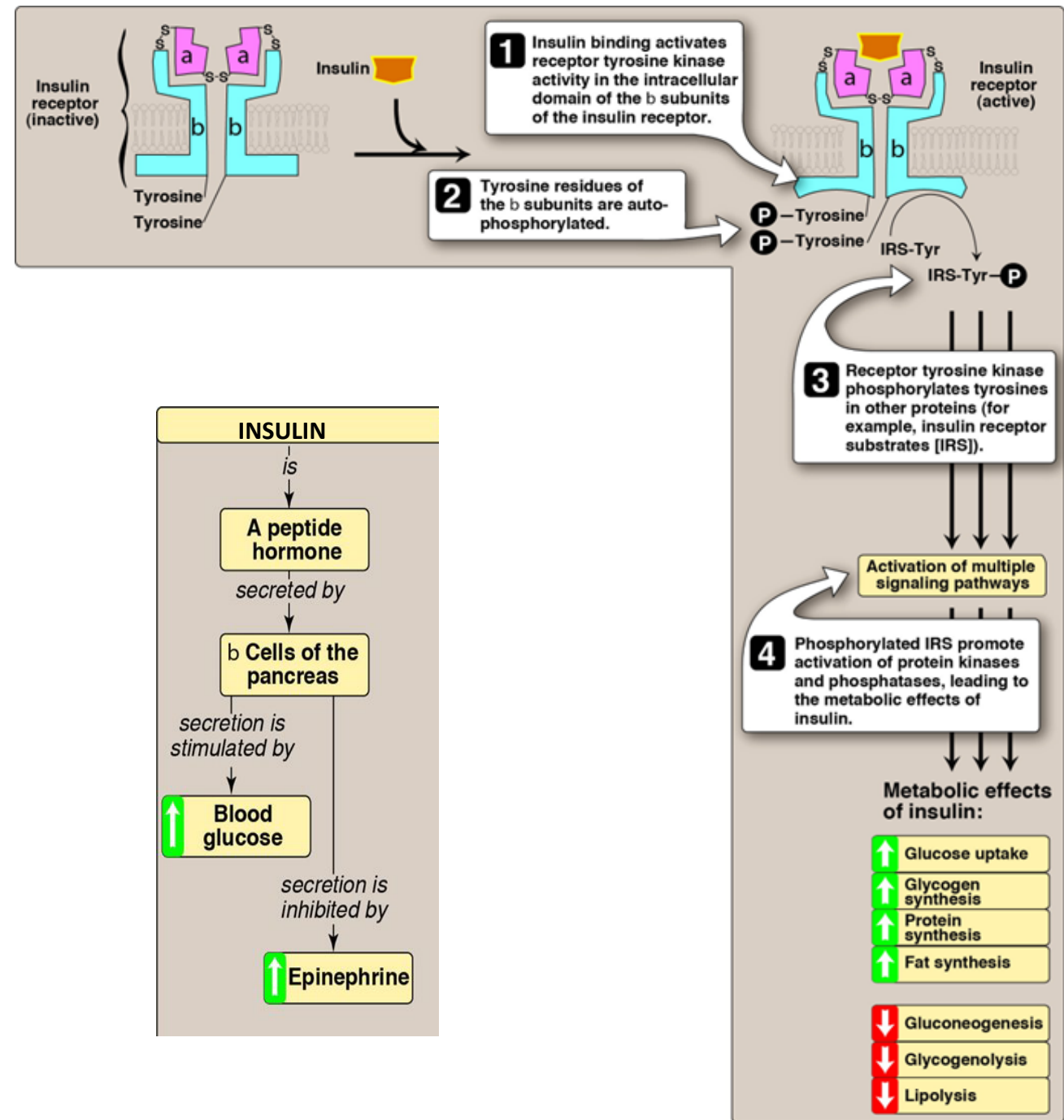


Normal Insulin Actions

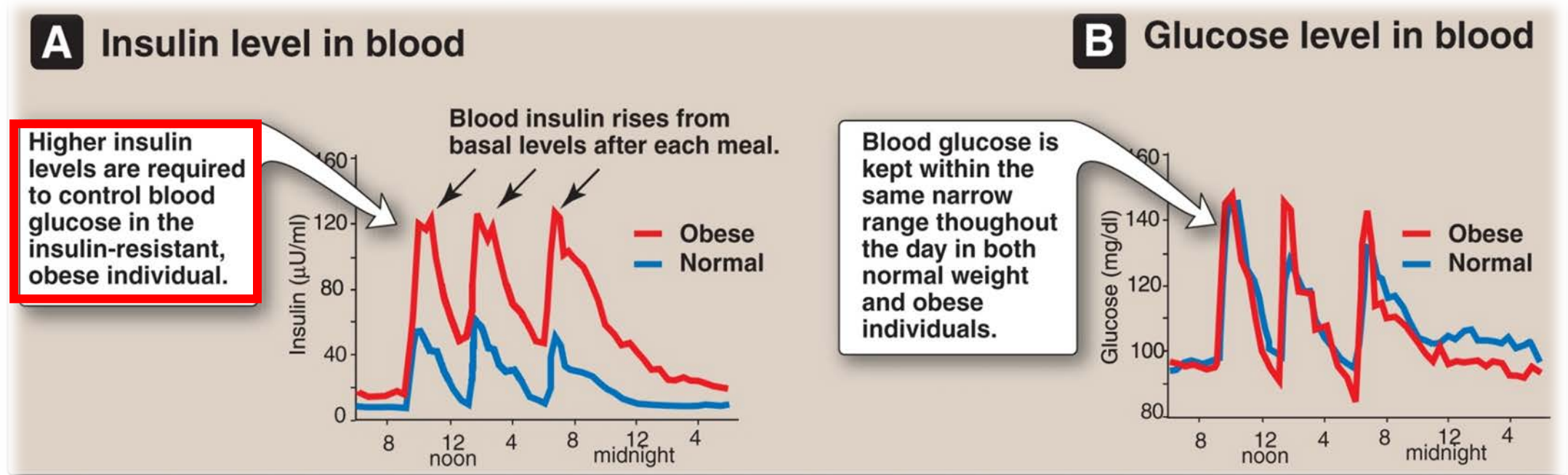
- ✓ Stimulates lipogenesis,
- ✓ Increases amino acid uptake/protein synthesis
- ✓ Increases glycogen storage
- ✓ Inhibits lipolysis,
- ✓ Inhibits gluconeogenesis and
- ✓ Inhibits glycogenolysis

Insulin Resistance

- ✓ Precedes development of type 2 diabetes mellitus (prediabetes)
- ✓ Risk factors: Obesity and sedentary lifestyle
- ✓ Increased insulin secretion



Insulin Resistance and Obesity



- **Strong correlation** between weight gain and development of insulin resistance. i.e. weight gain increases insulin resistance and weight loss decreases insulin resistance (improves insulin sensitivity)
- **Free fatty acids, leptin, and low adiponectin and cytokines** produced by **adipocytes** contribute to development of **insulin resistance**.

Insulin Resistance

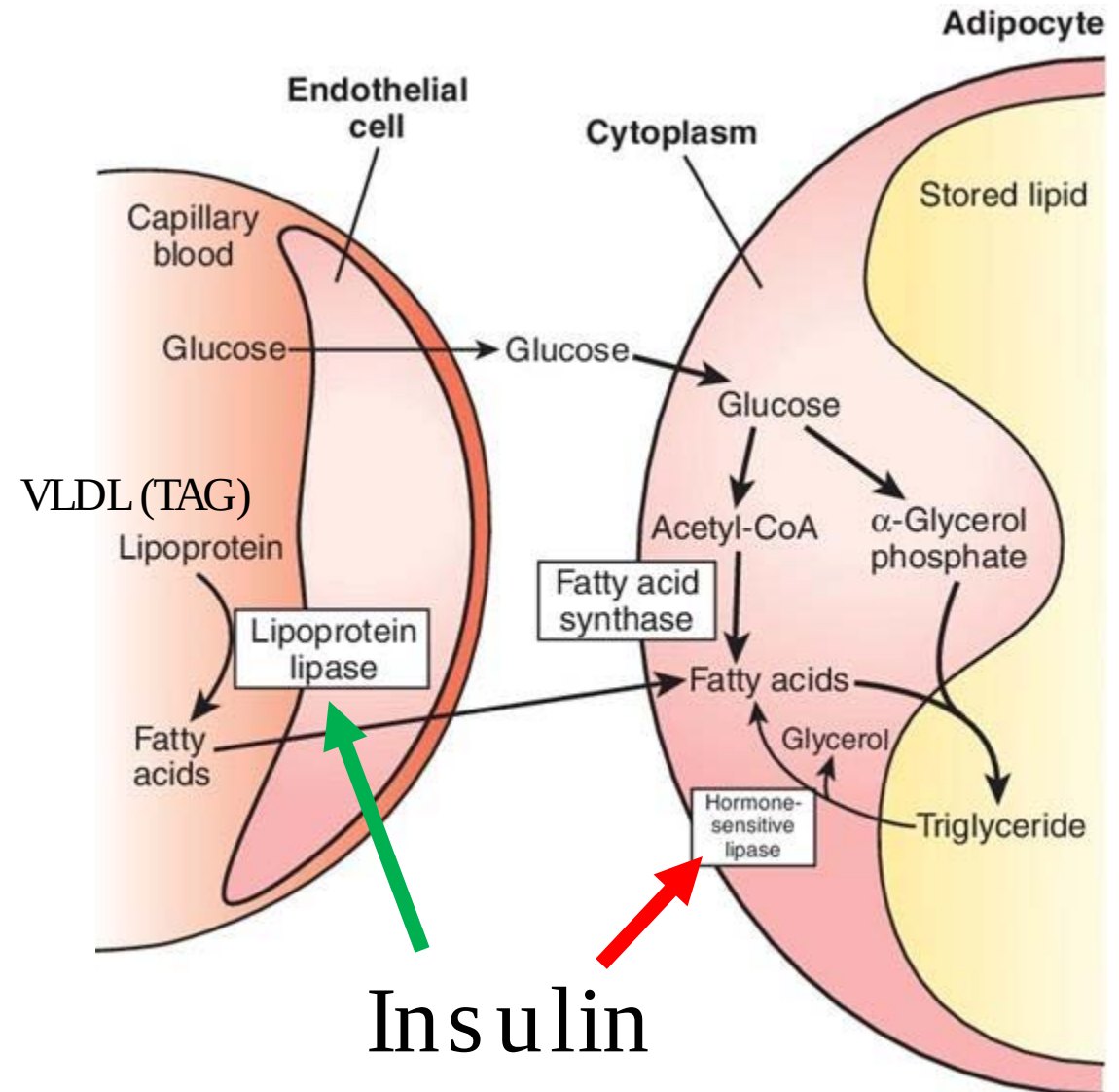
- Decreased number of insulin receptors
- Decreased insulin receptor (tyrosine kinase) activity
- Complex changes in intracellular metabolic pathways resulting in target tissue insensitivity to insulin
- Insulin resistance (Metabolic syndrome)
- **Dyslipidemia**
 - Increased serum triglycerides (TAG)
 - Due to increased synthesis of VLDL by liver and
 - **Low endothelial lipoprotein lipase activity**
 - Low HDL-cholesterol
 - Elevated LDL-cholesterol
 - Results in **increased atherosclerosis risk**



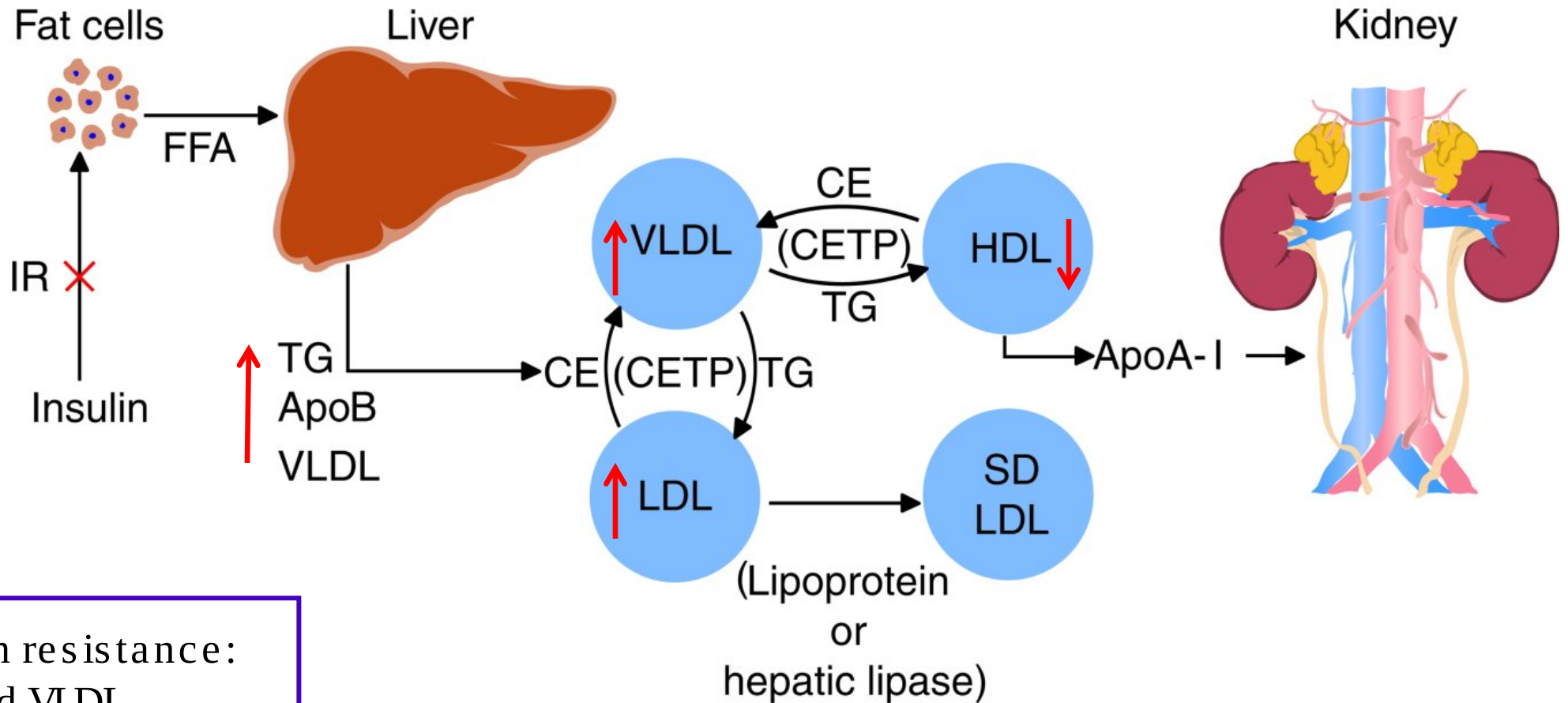
Acanthosis nigricans: Dark thick skin on neck, armpits or groin

Metabolic Changes Associated with Insulin Resistance

- Decreased **endothelial lipoprotein lipase** leading to increased serum triglycerides (TAGs) (mainly VLDL)
- Increased lipolysis (*hormone sensitive lipase*) in **adipose tissue**
 - Increased fatty acids in plasma
- Increased glucose output by **liver** and decreased GLUT-4 → Increased plasma glucose
- Increased protein degradation in **muscle**

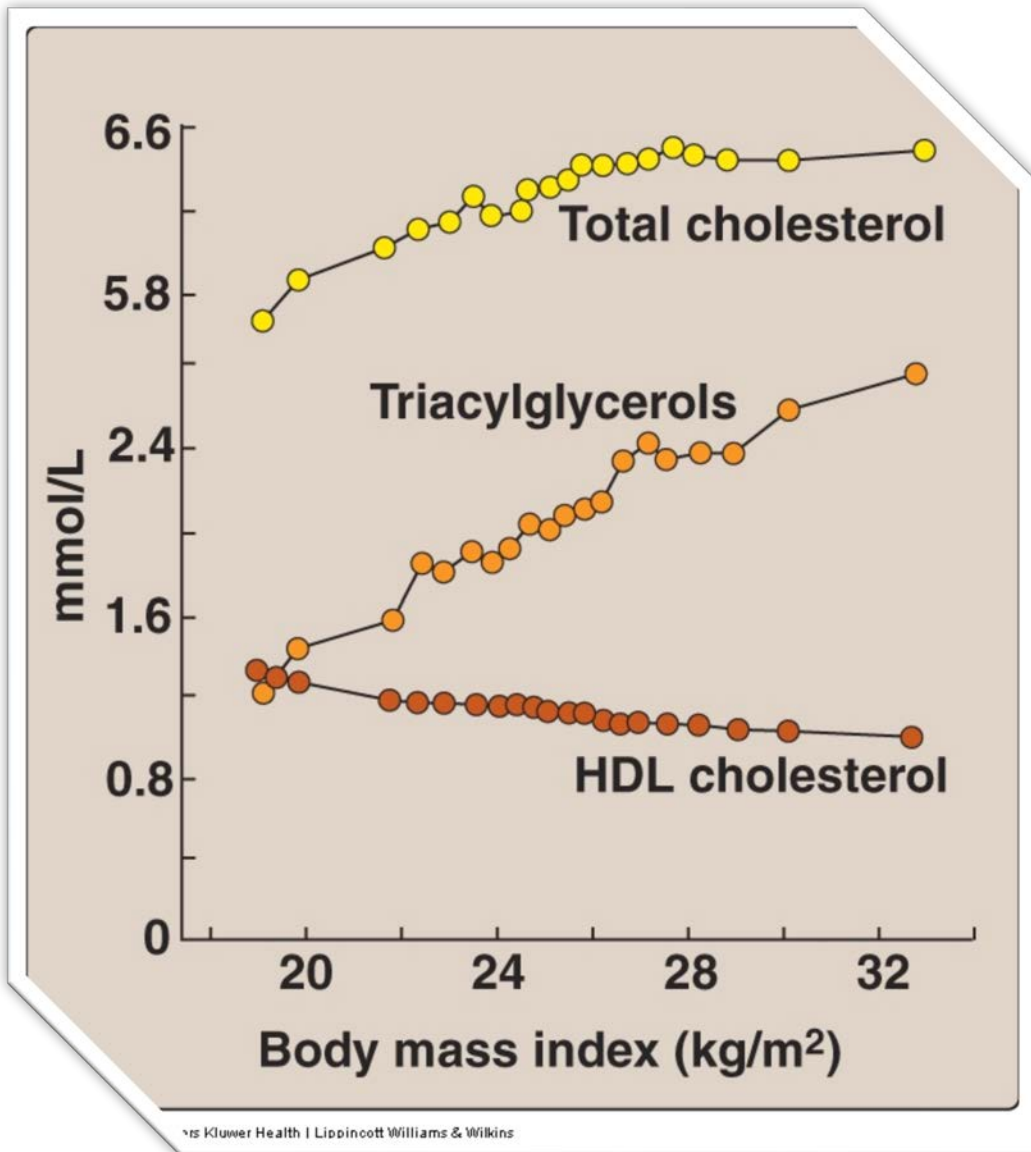


Effects of Insulin Resistance on Lipid Metabolism



CETP = Cholesteryl ester transfer protein; CE = Cholesteryl Ester; SD LDL = small dense LDL cholesterol; HL = hepatic lipase; FFA = free fatty acids

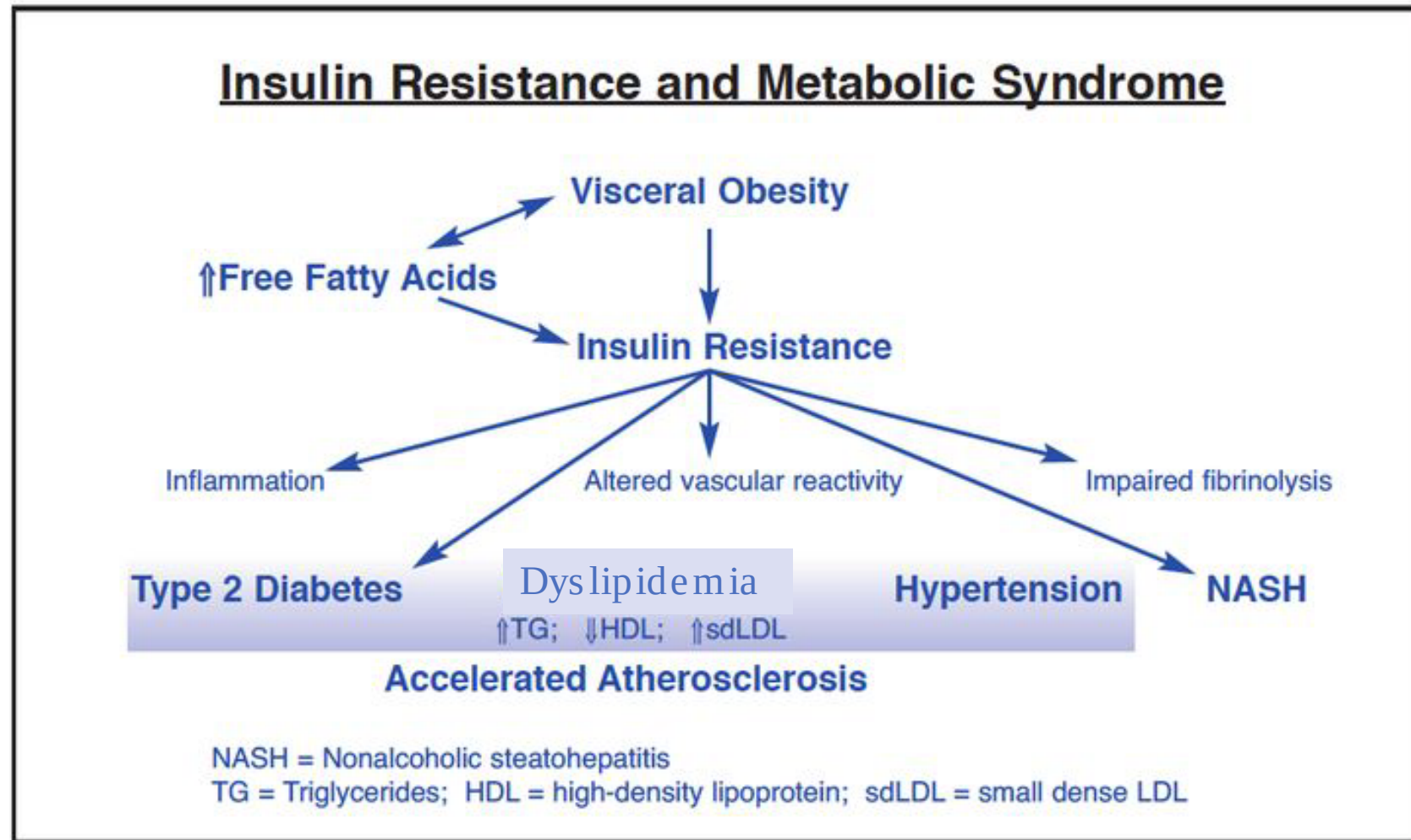
Obesity and Serum Lipid profile changes



- Elevated serum triglyceride (TAG/TG)
- Elevated total serum cholesterol
- Higher LDL-cholesterol
- Lower HDL-cholesterol
- **Atherogenic profile**

Obesity, Insulin Resistance and Metabolic Syndrome

- Ineffective insulin action in target tissues and compensatory hyperinsulinemia → Cardiovascular risk factors (Obese, sedentary, and diabetes)
- **Metabolic syndrome:** Common Metabolic abnormalities in these individuals



Metabolic Syndrome (Syndrome X)

- Clustering of atherosclerotic and cardiovascular disease **risk factors** prevalent in obese, sedentary, and/or diabetic individuals
- **Clinical signs**
 - Central (upper body) obesity with increased waist circumference
 - Acanthosis nigricans (hypertrophic, hyperpigmented skin changes)
- **Laboratory Abnormalities**
 - Elevated fasting and/or postprandial blood glucose/ HbA1c
 - Insulin resistance with hyperinsulinemia
 - Hypertension (endothelial and vascular dysfunction)
 - Dyslipidemia: Elevated triglycerides; Low HDL-cholesterol
 - Hypercoagulability and abnormal thrombolysis
 - Hyperuricemia → Gout
- **Comorbidities:** Hypertension, Atherosclerosis, Hyperandrogenism with polycystic ovary syndrome.

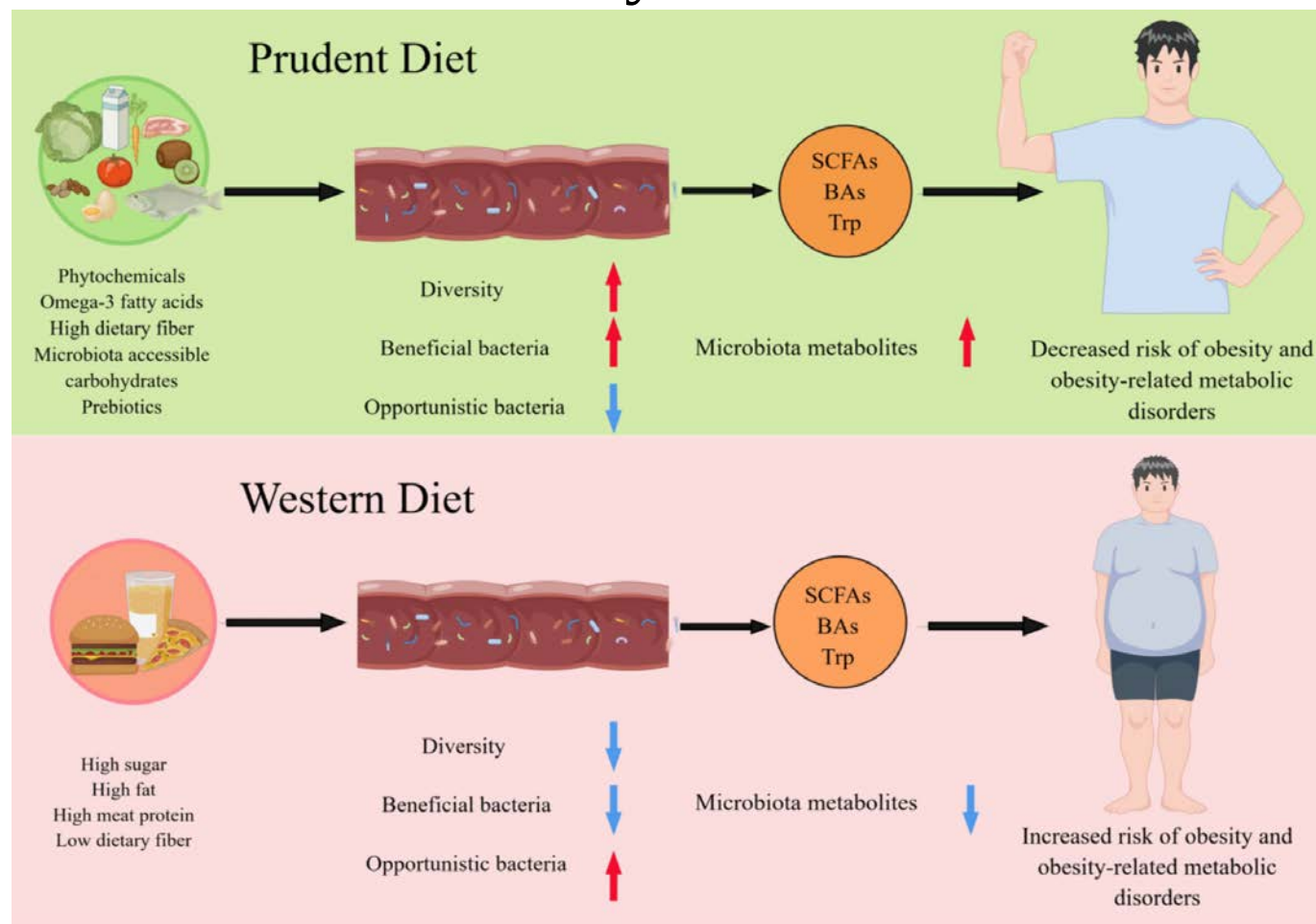
Metabolic Syndrome (presence of 3 or more criteria)

- Metabolic syndrome increases risk of cardiovascular disease, stroke, and Type-2 diabetes mellitus
- Presence of at least 3 out of 5 criteria in the table
- Management: Lifestyle changes and/or medication

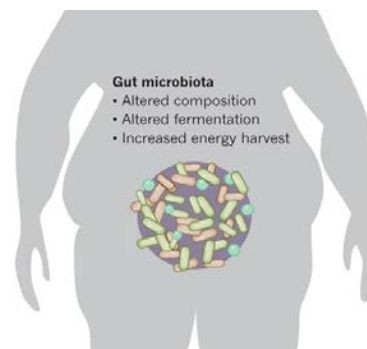
Criteria	Value
Elevated fasting blood glucose	≥ 100mg/dL
Elevated waist circumference (Central obesity)	≥ 40 inches men; ≥ 35 inches women
Elevated triglycerides (TAG)	≥ 150mg/dL
Reduced HDL cholesterol	< 40mg/dL men; < 50mg/dL women
Elevated blood pressure	≥ 130/80mm Hg

Gut microbiome: Correlation to Diet and Obesity

- Calorie-dense foods contain simple carbohydrates and oils → Readily absorbed without microbial assistance
- Gut microbiota affected by composition of diet
- Affects adiposity and response to diet
- Distal gut flora in obese people contain different microbes compared to flora of lean individuals
- Gut bacteria in obese individuals may extract calories from complex components of diet more efficiently than microbiota from lean counterparts.
- Regulatory effects on host from gut flora variations are being investigated, and gut flora “transplants” from lean to obese subjects have been proposed.



Gut Microbiome: Effects of diet



Features of the gut microbiota that promote obesity and insulin resistance: <https://www.nature.com/articles/nature11552>

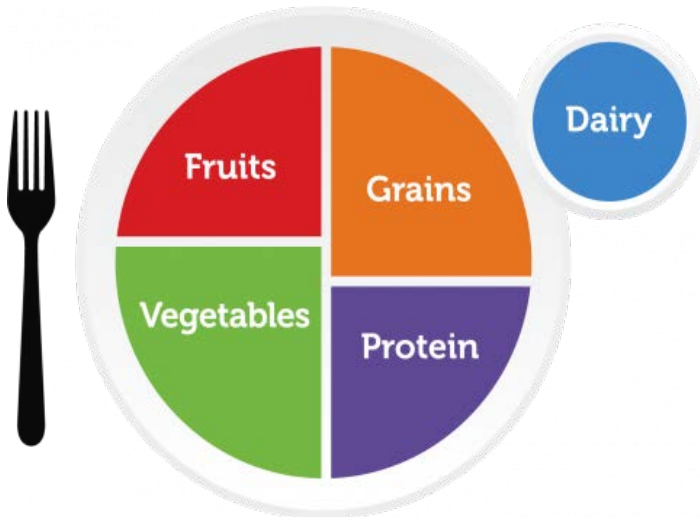
Managing Obesity

1. Lifestyle modifications:

- Decrease caloric intake: limit to 800-1000 kcal/day
- Increase physical activity
- Improve quality of food
 - Less saturated fat, sodium, and refined sugar
 - Increase fruit and vegetables,
 - Increase dietary fiber

2. Pharmacologic agents:

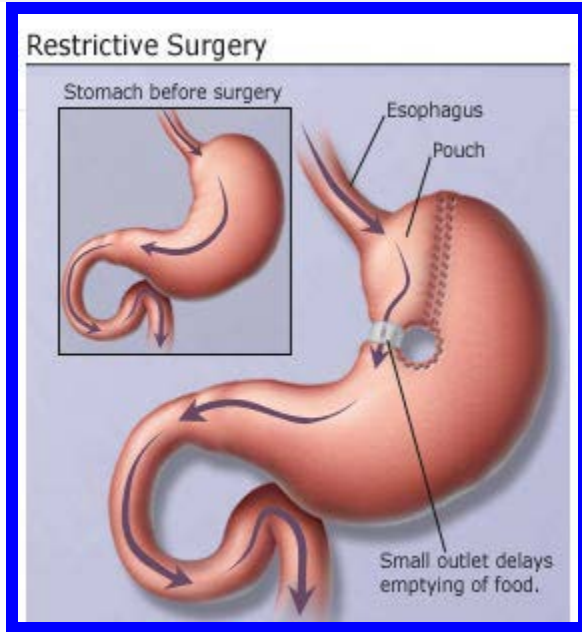
- **GLP-1 (Incretin):** Stimulates insulin secretion
- **Sibutramine:** Appetite suppressant
 - Increases fullness (satiety)
- **Orlistat:** Gastric & Pancreatic lipase inhibitor,
 - Inhibits the digestion of dietary fats (TAG's) and
 - Reduces dietary TAG absorption.



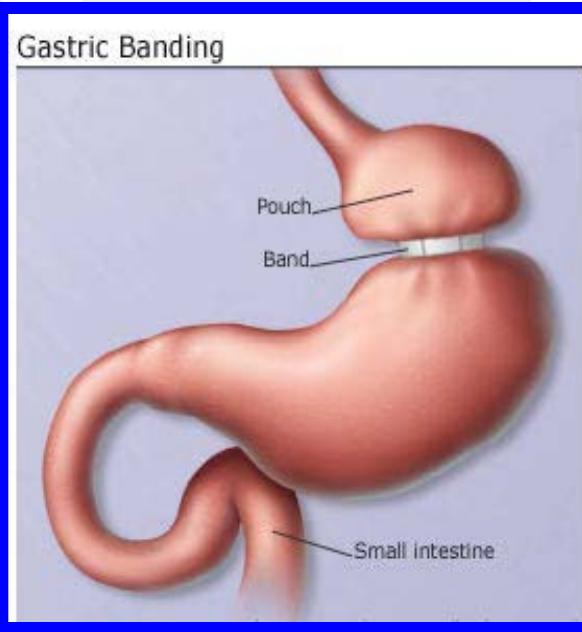
3. Surgical management:

- ✓ Recommended for morbidly obese (**BMI>35**).
- ✓ Intended to reduce food intake (smaller portions) and/or
- ✓ To reduce overall food absorption
- ✓ In conjunction with lifestyle modifications to achieve long-term weight loss

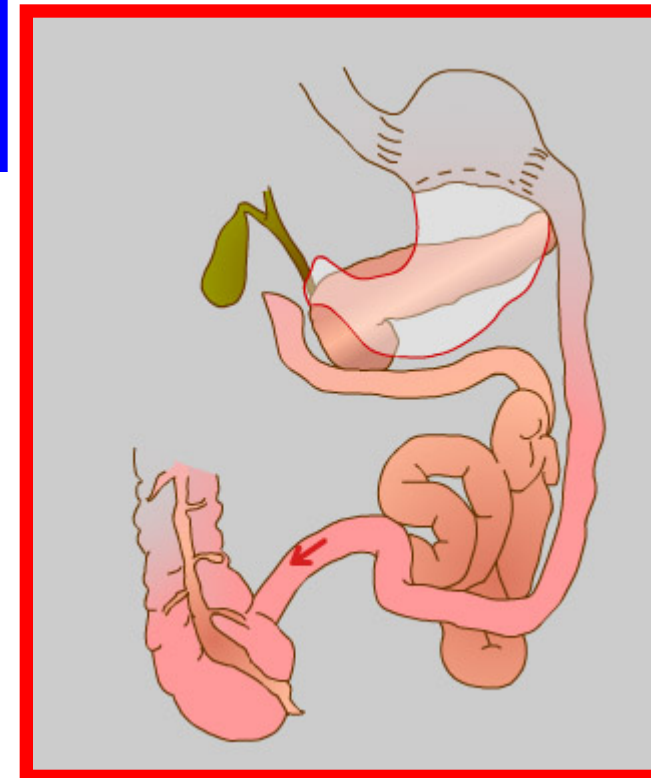
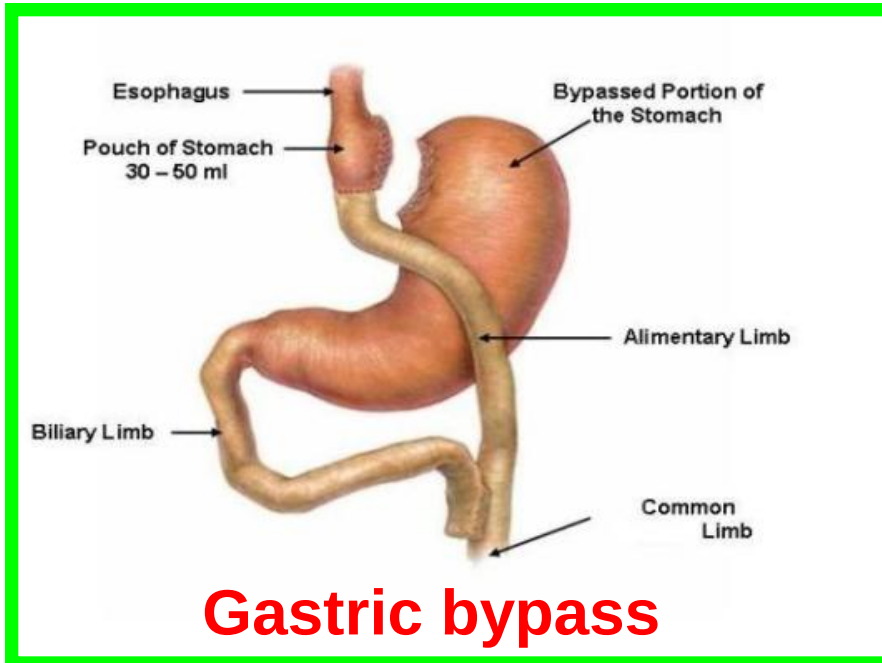
Reconstructive Surgery



G-banding



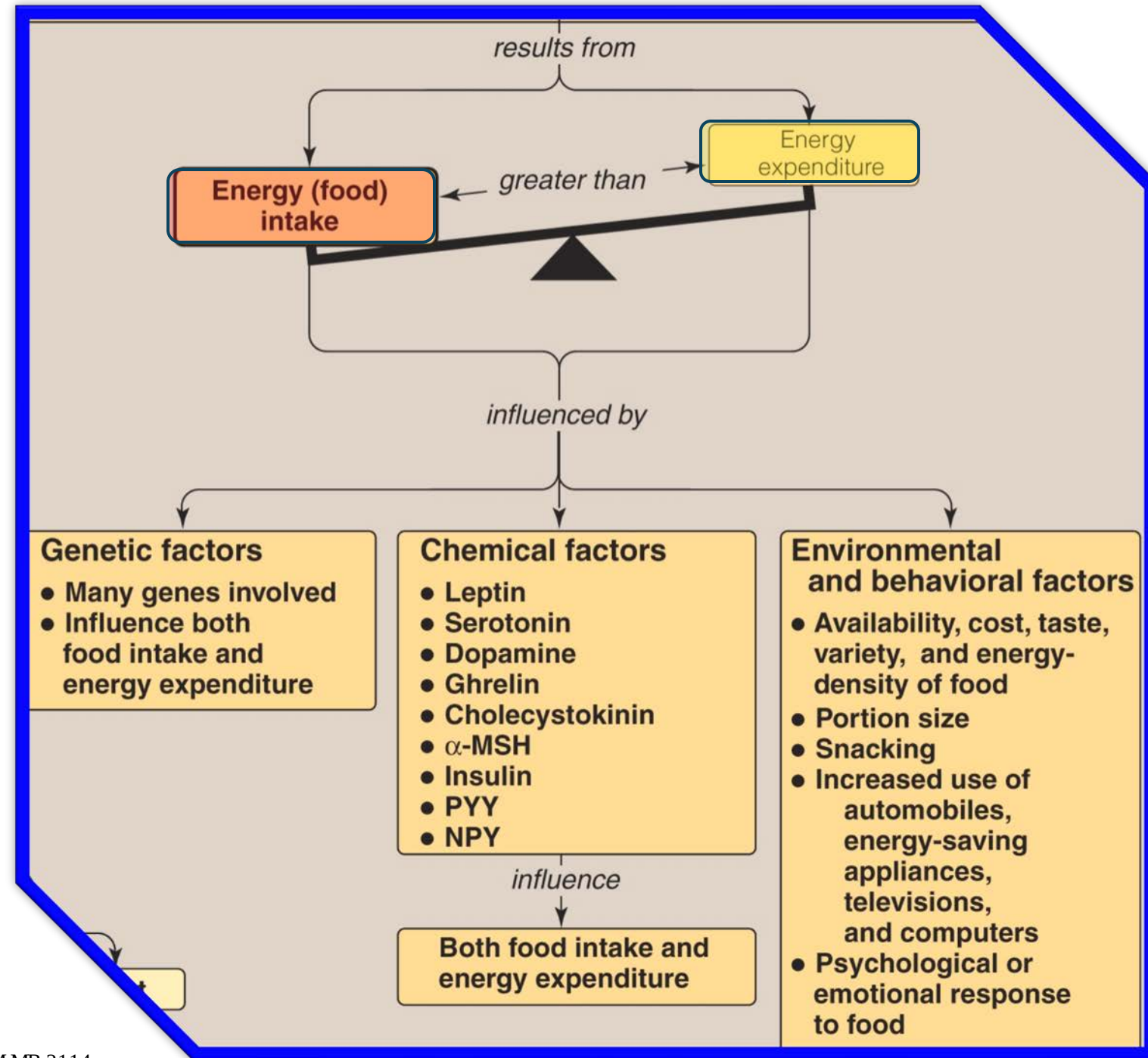
Surgical Procedures For Treating Obesity



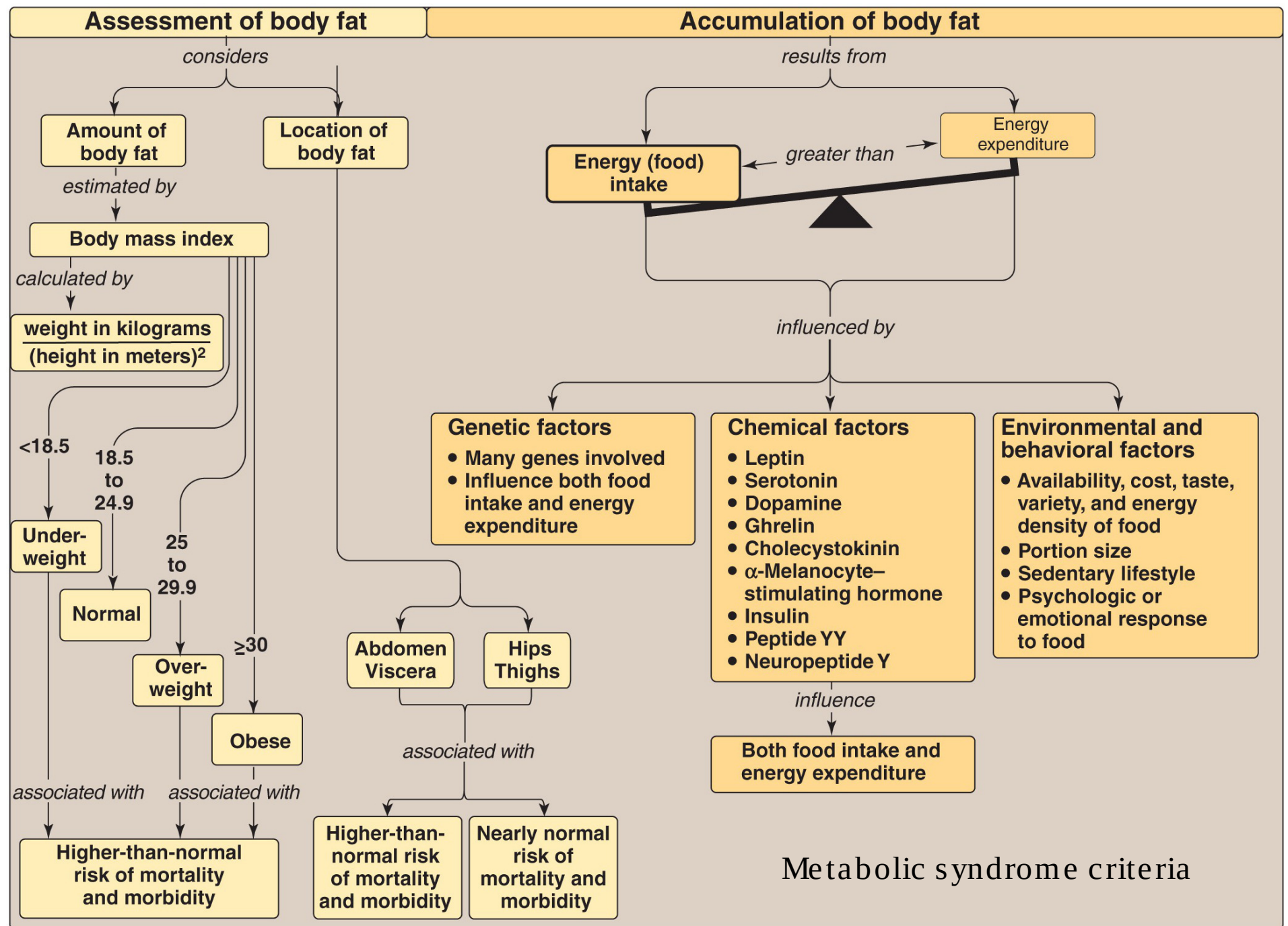
Ileo-gastric bypass

Ultimately, weight gain results from an imbalance between energy intake and energy expenditure.

Regulation involves genetic, hormonal, behavioral, and environmental factors.



Summary



Thank you!

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